Ahmad Fahadh Ilyas

AI Researcher
Jakarta / Banten, Indonesia · +62 813-1844-7472 · hello@fahadh.dev
fahadh.dev | github.com/fahadh4ilyas | linkedin.com/in/fahadh4ilyas

Summary

Born 19 April 1996 in Jakarta. Graduated from Universitas Indonesia in 2018 with a Bachelor's degree in Computational Mathematics. AI Researcher at PT eBdesk Teknologi working with LLMs, transformers, and speech synthesis. Active open-source contributor — patches and improvements to vLLM, Transformers, and diffusers, focused on model inference, quantization, and efficient fine-tuning. Author of a 20-page Kubernetes documentation guide in Bahasa Indonesia. Fast learner with the ability to self-teach effectively. Confident in English both written and spoken; Bahasa Indonesia native.

Work Experience

AI Researcher

PT eBdesk Teknologi — *Full-time*
Feb 2023 – Present (3 yrs 5 mos)
South Tangerang, Banten, Indonesia · On-site

- Research and development of AI/ML solutions with focus on LLMs, transformers, and speech synthesis (TTS)
- Model inference optimization, quantization, and efficient fine-tuning
- Open-source contributions around vLLM, Transformers, and diffusers

Back End Developer

PT eBdesk Teknologi — *Full-time*
Dec 2019 – Feb 2023 (3 yrs 3 mos)
Jakarta Metropolitan Area

- Developed back-end services and APIs
- Built smart contract boilerplates and Ethereum client tooling (web3py, py-solc-x)
- Worked with Blockchain and Python

Data Scientist

PT eBdesk Teknologi — *Full-time*
Mar 2019 – Dec 2019 (10 mos)
Jakarta Metropolitan Area

- Built end-to-end machine learning pipelines
- Create dataset and train machine learning models using Scikit-learn library
- Analyze the model and fine-tune it to increase certain metrics

Staff Programmer

PT Indonesia Indicator — *Full-time*
Mar 2018 – Mar 2019 (1 yr 1 mo)
South Tangerang, Indonesia

- Python development and software engineering

Education

University of Indonesia

Bachelor's Degree, Mathematics
Sep 2014 – Jan 2018 (3 yrs 4 mos)
GPA 3.86

Skills

AI / Machine Learning

LLMs, transformers, vLLM, PyTorch, diffusers, TTS (text-to-speech), speech synthesis, model inference, model quantization, efficient fine-tuning, DeepSeek, TensorFlow, Hugging Face ecosystem

Computer Vision

YOLO, real-time object detection, GPU-accelerated image processing, TensorRT, ESRGAN (super-resolution), MJPEG streaming, WebSocket, WebRTC

Backend & Infrastructure

Docker, Kubernetes (authored 20-page documentation guide in Bahasa Indonesia), Flask, REST APIs, WebSocket servers, microservices

Programming Languages

Python (primary), C++, JavaScript, Solidity (smart contracts)

Blockchain

Ethereum, web3py, py-solc-x, smart contract development

NLP / Linguistics

Indonesian syllable splitting, URL regex parsing

Other

GIS / GeoJSON (Indonesia geographic data), amateur radio software (WSJT-X), Telegram bot development, technical documentation writing

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Projects

Image-Upscaler

GPU-accelerated image upscaling REST API built with FastAPI and TensorRT, featuring 14 ESRGAN models across 1× to 4× scale factors, automatic ONNX-to-TensorRT engine compilation, resolution presets up to 4K, and containerized deployment with NVIDIA Docker runtime

YOLO-Service

Real-time YOLO object detection streaming server built with FastAPI, serving annotated video over HTTP MJPEG, WebSocket, and WebRTC with a shared-inference architecture that routes a single capture thread to all concurrent clients. Supports dual backends (Ultralytics PyTorch and custom TensorRT engine), per-client resolution and frame-skip controls, and Docker deployment

VLLM-Forwarder

FastAPI-based reverse proxy that fronts multiple vLLM inference servers behind a unified OpenAI-compatible API, dynamically routing chat completions, TTS, and arbitrary endpoints via a SQLite-backed model registry, with streaming support, WebSocket speech relay, structured JSONL logging, and containerized deployment

RealtimeRTC

OpenAI Realtime API-compatible voice conversation server delivering real-time speech-to-speech AI interactions over WebRTC, using SileroVAD, faster-whisper, and vLLM for the full VAD-STT-LLM-TTS pipeline with barge-in interruption, chain-of-thought reasoning display, and streaming tool calling

kubernetes-documentation

20-page Kubernetes documentation in Bahasa Indonesia, built as a single-page application with Tailwind CSS and vanilla JavaScript, spanning core concepts through cluster setup, pod controllers, storage, and access control — deployed on GitHub Pages for the Indonesian developer community

llm-documentation

Comprehensive 16-page LLM guide in Bahasa Indonesia spanning the full lifecycle: hardware specs, NVIDIA driver & CUDA setup, Docker & Anaconda configuration, model loading from Hugging Face, vLLM deployment with OpenAI-compatible API interaction, and supervised fine-tuning via OpenChat with MLflow experiment tracking — delivered as a Tailwind CSS SPA with clean-path routing, per-page SEO metadata, dark/light theme, responsive sidebar, swipe gestures, and a contact form with Turnstile CAPTCHA

indonesia-geojson-archive

Curated repository of Indonesia's complete administrative boundary GeoJSON data spanning five levels — from country and province down to village — with corrections for accurate Indonesian regional naming, built on GADM source data (9 stars)

syllable_splitter

Rule-based Indonesian syllable tokenizer using a two-pass pattern-matching algorithm that classifies character sequences with language-specific digraphs (ng, ny, sy, kh) and applies phonotactic boundary rules via regex to split words and sentences into linguistically accurate syllable tokens (4 stars)

smart-contract-boilerplate

Ethereum smart contract boilerplate built on Truffle v5 and Solidity 0.8.x, integrating OpenZeppelin's audited contracts with upgradeable patterns and multi-network deployment across Ethereum testnets and Polygon Mumbai, with Ethlint, eth-gas-reporter, and Mocha-based testing via Ethers.js

Simple-Ethereum-Client

Python library wrapping Web3.py and py-solc-x into a single ETHClient class, streamlining account management, balance transfers, smart contract compilation and deployment, event log parsing, and transaction lifecycle control with built-in gas estimation, ABI parsing, and address checksumming

Auto-WSJT-X / **WSJT-X-2-6-0**

Automated FT8/FT4 digital-mode operator: modified the WSJT-X C++/Qt application to expose extended UDP control messages, then built a Python automation framework that parses the WSJT-X network protocol, maintains state via Redis and MongoDB, intelligently selects contacts through DXCC prioritization and frequency-gap analysis, and completes fully automated two-way QSOs with logging

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Languages

- **Bahasa Indonesia** — Native
- **English** — Limited working proficiency

Profiles

- **Email:** hello@fahadh.dev
- **Phone:** +62 813-1844-7472
- **Website:** fahadh.dev
- **GitHub:** github.com/fahadh4ilyas
- **LinkedIn:** linkedin.com/in/fahadh4ilyas